INTELLIGENT SETTLEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfillment of requirements for the award of the course

of

CGB1201 – JAVA PROGRAMMING

in

B.Tech - INFORMATION TECHNOLOGY

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(An Autonomous Institution, affiliated to Anna University Chennai and Approved by AICTE, New Delhi)

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NOVEMBER – 2026

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Under
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BONAFIDE CERTIFICATE

Certified that this project report titled **"INTELLIGENT SETTLEMENT SYSTEM"** is the bonafide work of **ALAGAPPAN K (8115U25IT005)** and **KEVIN XAVIER (8115U25IT056)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate during the academic year 2026-2027.

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We declare that the project report on **"INTELLIGENT SETTLEMENT SYSTEM"** is the result of original work done by us and best of our knowledge, similar work has not been submitted to **"ANNA UNIVERSITY CHENNAI"** for the requirement of Degree of BACHELOR OF TECHNOLOGY. This project report is submitted on the partial fulfillment of the requirement of the award of the course **CGB1201- JAVA PROGRAMMING** during the academic year 2026-2027.

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ACKNOWLEDGEMENT

It is with great pride that we express our gratitude and indebtedness to our institution, **"K.RAMAKRISHNAN COLLEGE OF ENGINEERING (Autonomous)"**, for providing us with the opportunity to do this project. We extend our sincere acknowledgment and appreciation to the esteemed and honorable Chairman **Dr. K. RAMAKRISHNAN, B.E.,** for having provided the facilities during the course of our study in college.

We would like to express our sincere thanks to our beloved Executive Director, **Dr. S. KUPPUSAMY, MBA, Ph.D.,** for forwarding our project and offering an adequate duration to complete it.

We would like to thank **Dr. D. SRINIVASAN, M.E., Ph.D., FIE., MIIW., MISTE., MISAE., C. Engg.,** Principal, who gave the opportunity to frame the project to full satisfaction.

We thank **Dr. S. MANIKANDAN, M.E., Ph.D.,** Head of the Department of Information Technology, for providing his encouragement throughout the project.

We wish to convey our profound and heartfelt gratitude to our esteemed project guide **Mrs. S. SATHYA PRIYA, M.E., Ph.D.,** Department of Information Technology, for her incalculable suggestions, creativity, assistance and patience, which motivated us to carry out this project.

We render our sincere thanks to the staff members for providing valuable information during the project work.

INSTITUTE VISION AND MISSION

VISION OF THE INSTITUTE

To achieve a prominent position among the top technical institutions

MISSION OF THE INSTITUTE

M1: To bestow standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.

M2: To nurture research and entrepreneurial skills among students in cutting edge technologies.

M3: To provide education for developing high-quality professionals to transform the society.

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VISION OF THE DEPARTMENT

To create technically skilled IT professional to meet corporate expectations and addressing societal concerns.

MISSION OF THE DEPARTMENT

M1: To build competency in advanced computing technologies and implement project based learning.

M2: To establish industry collaboration for research projects and consultancy.

M3: To inculcate entrepreneurial skills to serve the societal benefits.

PROGRAM EDUCATIONAL OBJECTIVES(PEOs)

PEO1: Solve real world problems using advanced computing technologies.

PEO2: Encompass the ability to build innovative software products and become a successful entrepreneur.

PEO3: Carryout industrial projects and succeed in research careers.

PROGRAM OUTCOMES

PROGRAM OUTCOMES(POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization as specialization WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society, and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.(WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES(PSOs)

PSO1: Apply the knowledge of computer programming to develop the software projects.

PSO2: Design software in futuristic approach to adapt cutting edge technologies.

PSO3: Solve societal problems using computation intelligence.

ABSTRACT

The *Intelligent Settlement System* is a user-centric web platform designed to simplify the management of shared expenses and the settlement of outstanding balances among groups. When friends, roommates, classmates, or colleagues share payments, it becomes difficult to calculate each person's contribution and to decide who should pay whom. The system provides an organized environment where users can create groups, add members, record expenses, and view the resulting balances in a single workflow. Each expense is recorded with its payer, participants, amount, date, and category, and the individual shares are calculated using equal, percentage-based, or custom splitting methods. After the shares are calculated, the system determines the net balance of every member by comparing the amount paid with the amount owed. A key feature of the system is the debt simplification engine, which matches members who owe money with members who are owed money and recommends a reduced set of settlement transactions instead of requiring every expense to be settled separately. The expense history module maintains a complete record of all transactions for future reference, while the settlement and reporting module presents suggested payments, pending dues, and a consolidated final expense and settlement report. The administrative oversight module allows groups and transactions to be monitored. The website is developed using HTML, CSS, and JavaScript for the front end, PHP for the back end, and MySQL for data storage. The overall system reduces manual calculation, avoids confusion between multiple dues, and makes group financial coordination transparent, accurate, and easy to understand.

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LIST OF ABBREVIATIONS

ACRONYM	ABBREVIATION
API	Application Programming Interface
CSS	Cascading Style Sheets
DBMS	Database Management System
GB	Gigabyte
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment
JS	JavaScript
PHP	Hypertext Preprocessor
RAM	Random Access Memory
SQL	Structured Query Language
UI	User Interface
UX	User Experience
XAMPP	Cross-Platform, Apache, MySQL, PHP, Perl

CHAPTER 1

INTRODUCTION

1.1 OBJECTIVE

The objective of the Intelligent Settlement System is to develop an efficient and organized platform for managing shared expenses and settling outstanding balances among groups. The system aims to automate processes such as group and member management, expense recording, share calculation, balance tracking, and settlement planning, thereby reducing manual effort and minimizing calculation errors. It maintains centralized records of users, groups, expenses, shares, balances, and settlements, ensuring easy access and consistency. The system enhances user convenience by allowing members to record who paid, choose how the cost is shared, and immediately see who owes whom. It includes a debt simplification feature that recommends a smaller set of transfers to clear the group's dues. The system also provides an expense history and a consolidated final report for transparent financial coordination. Overall, it improves accuracy, saves time, and reduces confusion while supporting scalable group expense management.

1.2 OVERVIEW

The Intelligent Settlement System is developed to provide a structured and organized solution for managing shared costs efficiently. In everyday life, groups of friends, roommates, classmates, and colleagues frequently share expenses such as food, travel, accommodation, and events. One member often pays on behalf of the others, and different members pay for different expenses over time. Tracking each payment manually through notes or messages results in repeated calculations, unclear balances, and unnecessary back-and-forth transfers. Therefore, there is a strong need for a centralized system that can manage group expenses in a clear and systematic manner.

The system serves as a centralized platform where all group and expense data is stored and maintained in an organized structure. Users first create a group and add its members. Each expense is then entered along with its amount, the member who paid, and the members who share it. The expense splitting component calculates each participant's share using equal, percentage-based, or custom methods, and the balance calculation combines these shares with the amounts already paid to produce a net balance for every member.

The project focuses on simplifying settlement by creating a platform that supports easy access to services such as group management, expense entry, expense history, debt simplification, and reporting. The debt simplification component matches members who owe money with members who are owed money and suggests a reduced set of payments. The settlement and reporting section then presents the suggested transfers, the pending dues, and the overall financial status of the group. Clear information about every expense and balance is displayed at all times, which builds trust among the members. The system is designed to be simple, accurate, and user-friendly, and it can be extended with additional features in the future.

CHAPTER 2

LITERATURE SURVEY

2.1 EVOLUTION OF SHARED EXPENSE MANAGEMENT

Sharing costs has always been a part of social and domestic life, but the way in which shared expenses are recorded has changed considerably. Earlier, groups relied on handwritten notebooks and verbal agreements to remember who had paid for what. These informal methods worked for very small groups, but they depended entirely on memory and trust, and disagreements were common when records were incomplete or unclear.

With the spread of computers and smartphones, spreadsheets and messaging groups became the common tools for tracking shared spending. Spreadsheets allowed totals to be calculated automatically, yet they still required manual entry of every share and manual comparison of what each person had paid and owed. As group activities such as shared housing, student trips, and team events became more frequent, the need for dedicated tools that could record expenses and compute balances automatically became evident.

Despite the availability of several tools, many solutions remain limited to recording expenses and do not help the group decide how to settle them with the fewest payments. This gap highlights the need for an integrated system that combines expense recording, share calculation, balance tracking, and settlement planning in a single platform.

2.2 EXISTING EXPENSE-SHARING APPLICATIONS

A number of mobile and web applications are available for splitting bills among friends and groups, and applications such as Splitwise are widely used for this purpose. These applications allow users to add expenses, divide them among

members, and view the amount that each member owes or is owed. They have made shared expense tracking far more convenient than manual methods.

However, general-purpose applications are designed for very broad use and often include features that are unnecessary for small academic or project-based groups, while lacking the flexibility to be adapted to a specific institution or workflow. Many of them also require users to register through external accounts, rely on proprietary servers, and offer limited control over how data is stored and reported.

Therefore, a purpose-built system that can be hosted locally, customized, and explained clearly is valuable for learning and for small communities. Such a system should demonstrate the complete workflow from group creation to final settlement, using transparent logic that users can understand and verify.

2.3 EXPENSE SPLITTING METHODS AND FAIRNESS

The way in which an expense is divided among participants determines whether the outcome is perceived as fair. The simplest approach is the equal split, in which the total amount is divided evenly among all selected participants. This method is quick and suitable when everyone benefits equally from the expense, such as a shared meal or a common taxi ride.

In many situations, however, members do not consume equally. Percentage-based splitting allows each participant to be assigned a proportion of the total, while custom splitting allows an exact amount to be entered for each person. These methods require validation so that the percentages add up to one hundred percent and the custom amounts add up to the total expense. Without such checks, the recorded shares may not match the amount actually paid, leading to incorrect balances.

Existing tools often provide only one or two splitting options, which limits their usefulness. A system that supports equal, percentage-based, and custom splitting within the same interface allows groups to record real-life situations accurately and to maintain trust among members.

2.4 NET BALANCE AND DEBT SIMPLIFICATION ALGORITHMS

Once all shares are known, the financial position of each member can be summarized as a net balance, which is the difference between the amount the member has paid and the amount the member is responsible for. A positive net balance means the member should receive money, and a negative net balance means the member should pay. In a correctly recorded group, the sum of all net balances is zero.

Settling every expense individually can create a large number of small transfers. Debt simplification reduces this effort by working only with net balances. A commonly used greedy approach repeatedly matches the member who owes the most with the member who is owed the most and transfers the smaller of the two amounts. Each transfer clears at least one member completely, so a group of n members with outstanding balances needs at most $n - 1$ transfers.

Finding the absolute minimum number of transfers for every possible case is known to be a computationally hard problem, so practical systems use efficient heuristics such as the greedy method. This produces a valid and understandable settlement plan for small and medium groups, although it does not guarantee a globally optimal solution in every case.

2.5 WEB TECHNOLOGIES AND SYSTEM ARCHITECTURE

Web technologies form the backbone of modern software applications by enabling the development of interactive and accessible systems. The front end is responsible for user interaction and is developed using HTML, CSS, and JavaScript,

which together create structured, styled, and responsive user interfaces that run in any standard browser without additional installation.

The back end handles server-side operations such as business logic, data processing, and communication with the database. Technologies such as PHP, Node.js, and Python are commonly used for this purpose due to their flexibility and wide community support. Relational databases such as MySQL are widely used for storing structured data such as users, groups, expenses, and settlement records, because they ensure integrity, consistency, and reliable retrieval of information.

The LAMP and XAMPP stacks, which combine Apache, MySQL, and PHP, are popular for developing and deploying web applications because they are reliable, cost-effective, and easy to configure. Despite these advantages, poor architectural design can lead to low performance and inefficient integration between components, which is why a clear modular architecture is important.

2.6 USER INTERFACE AND USER EXPERIENCE (UI/UX)

UI/UX design plays a vital role in determining the usability and success of a system by directly influencing how users interact with the application and perceive its functionality. A well-designed interface provides clear navigation, intuitive controls, and a visually organized layout, which helps users complete tasks quickly and with fewer mistakes.

Financial applications in particular benefit from interfaces that present figures clearly. Tables, summary cards, and colour-coded status labels help users understand amounts, balances, and pending dues at a glance. Responsive design principles ensure that the same pages remain usable on desktops, tablets, and smartphones, which is important because group members often enter expenses on the move.

User-centered design approaches focus on understanding user needs and behavior so that the interface is both functional and pleasant to use. For an expense

settlement system, a simple form for entering expenses and a clear summary of who owes whom are the most important interface elements.

2.7 DATA ACCURACY, VALIDATION AND ROUNDING

Accuracy is the most important quality of any system that handles money. Even a small error in a share calculation can make the total of the shares differ from the amount paid, and such differences accumulate over many expenses. For this reason, input validation is applied to confirm that amounts are positive numbers, that at least one participant is selected, and that percentages or custom shares add up correctly.

Rounding is another common source of discrepancy. When an amount cannot be divided exactly among the participants, for example when one hundred rupees is shared by three people, the shares cannot all be equal to the paisa. Reliable systems therefore store currency values with fixed precision and distribute any remaining paisa explicitly so that the shares always add up to the original amount.

Consistent validation and precision handling ensure that the sum of all net balances remains zero, which is the basic condition for a correct settlement. Many simple tools overlook these details, which leads to small but persistent balance errors.

2.8 SECURITY AND PRIVACY IN WEB SYSTEMS

Security is a fundamental requirement for any web application, especially one that handles personal and financial information, as it protects user data and maintains system integrity. An expense settlement system stores names, group memberships, and details of who owes money to whom, which should be visible only to the members concerned.

Common threats to web applications include SQL injection, cross-site scripting, and session hijacking. Commonly used protective techniques include validating and sanitizing all user input, using prepared statements for database

queries, storing passwords in hashed form, and controlling access through login sessions. Secure communication using HTTPS protects data while it is transmitted between the browser and the server.

Applying these measures reduces the chances of data leakage and unauthorized changes to financial records. A system that is designed with security in mind builds trust among its users and encourages them to rely on it for recording shared payments.

2.9 INTEGRATION CHALLENGES IN EXISTING SYSTEMS

One of the major limitations of current systems is the lack of integration between different functions. Expense recording, balance calculation, settlement planning, and reporting are often handled by separate tools or by separate manual steps. Users must then copy data from one place to another, which wastes time and increases the chance of errors.

This lack of coordination also affects decision-making, since complete and accurate information is not available in a single place. An integrated system provides a unified platform in which the output of each stage becomes the input of the next: recorded expenses lead to shares, shares lead to balances, balances lead to settlement suggestions, and settlements lead to updated reports.

Integration also ensures data consistency and better synchronization across all components, allowing balances to be updated immediately whenever a new expense or payment is recorded. This is a key requirement that the proposed system aims to fulfil.

2.10 EXISTING SYSTEM LIMITATIONS

Most existing approaches to shared expense management are developed to perform only specific tasks such as recording expenses or calculating a simple total, which limits their overall functionality. Manual methods depend on individual

memory and effort, while basic digital tools do not simplify debts and therefore leave the group with many separate payments to make.

Additionally, many applications do not provide a clear history of expenses or a consolidated report, making it difficult to review past transactions or to verify a balance when a disagreement arises. Some tools also have complex navigation or require unnecessary registration steps, which reduces their usability for small groups.

These limitations clearly highlight the need for a better system that is simple, integrated, accurate, and transparent, and that can record expenses, calculate shares, simplify debts, and report settlements within a single platform. The proposed Intelligent Settlement System is designed to meet this need.

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

3.1 EXISTING SYSTEM

The existing method of managing shared expenses is mostly manual or only partially computerized, making it inefficient and difficult to manage. Groups usually keep track of payments using notebooks, spreadsheets, or messaging applications, which leads to errors and data inconsistency. Members depend on memory to recall who paid for which expense, and the shares are calculated by hand for every purchase. When several members pay for different expenses, the group has no clear way to combine them, and the settlement is usually done expense by expense. This creates many unnecessary transfers and confusion over the pending amounts. Moreover, there is no proper record of past expenses or consolidated report, so verifying a balance later is difficult.

DEMERITS

- Time-consuming manual calculation of shares
- High chances of human error in balances
- Lack of a centralized record of expenses
- No instant update of member balances
- Many unnecessary transfers to settle dues
- Limited reporting and history of transactions

3.2 PROPOSED SYSTEM

The proposed Intelligent Settlement System is a fully integrated and automated platform designed to simplify the management of shared expenses. It provides a centralized system where users can create groups, add members, record

expenses, and choose a splitting method. The system calculates the individual shares automatically using equal, percentage-based, or custom splitting and updates the net balance of every member instantly. Its debt simplification engine combines these balances and suggests a reduced set of payments that clears all dues. The expense history module stores every record for future reference, and the settlement and reporting module presents suggested settlements, pending dues, and a final expense and settlement report. The system is developed as a user-friendly website, so that members can access it easily from a browser.

MERITS

- Centralized and integrated system
- Automatic calculation of shares and net balances
- Debt simplification with fewer transfers
- Instant balance updates after every expense
- Complete expense history and final report
- User-friendly web interface

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

Processor	:	Intel Core i3 or higher processor
RAM	:	Minimum 4 GB RAM
Hard Disk Drive	:	7200 RPM Hard Disk Drive
Hard Disk Space	:	Minimum 20 GB available space
Network	:	Stable Internet Connection

4.2 SOFTWARE REQUIREMENTS

Operating System	:	Windows 10 / Windows 11
Front End	:	HTML, CSS, JavaScript
Back End	:	PHP
Database	:	MySQL
Web Server	:	Apache (XAMPP / Laragon)
Web Browser	:	Chrome, Mozilla Firefox, Microsoft Edge
Development Tool	:	Visual Studio Code

CHAPTER 5

SYSTEM ARCHITECTURE

The architecture of the Intelligent Settlement System follows a structured and modular workflow designed to provide a seamless experience for managing shared expenses. The system is divided into multiple functional sections such as User and Group Management, Expense Entry, Expense Splitting, Debt Simplification, and Settlement and Reporting. Each section represents a specific stage in the life cycle of a shared expense, starting from user registration and group creation and ending with the final settlement report.

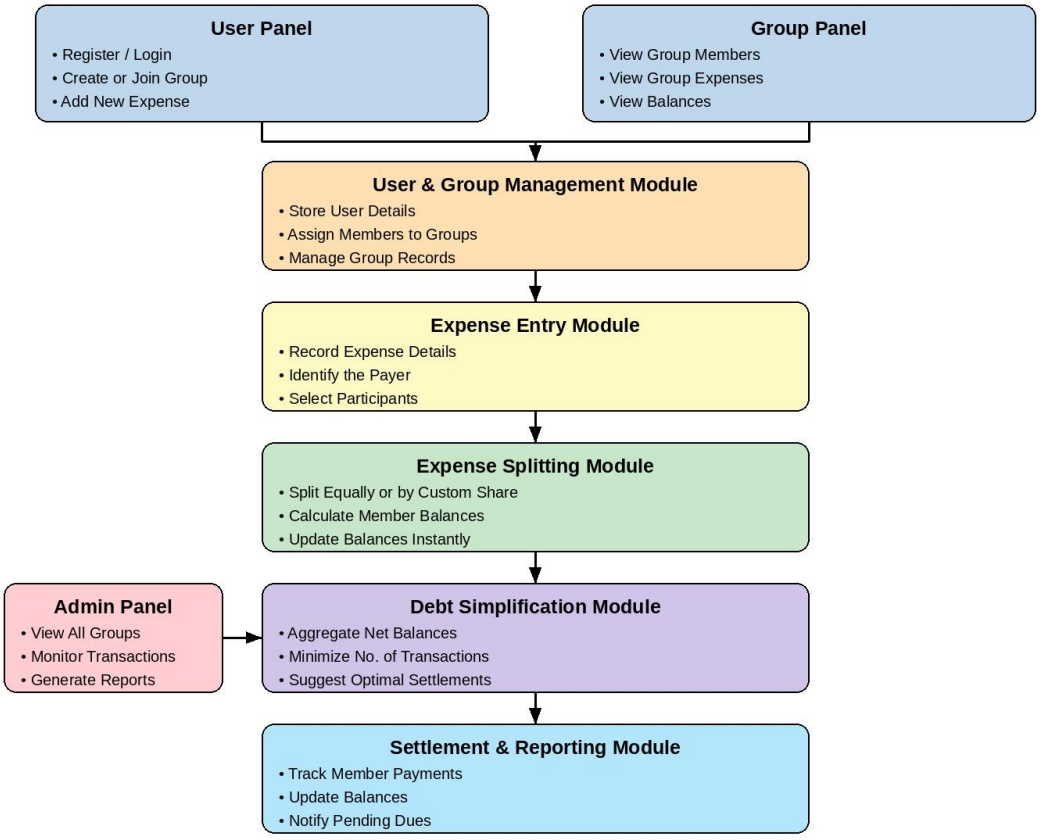


Fig.No. 5.1 System Architecture

At the top level, the system includes the user interface, which acts as the primary point of interaction between the users and the system. The User Panel allows a member to register or log in, create or join a group, and add a new expense,

while the Group Panel allows the members of a group to view the group members, the recorded expenses, and the current balances. Every action performed by the user is captured through web pages and sent to the application logic for processing, ensuring that all operations are handled systematically.

Each module performs a dedicated function and communicates with the next module when required. The User and Group Management Module stores user details and assigns members to groups. The Expense Entry Module records the expense details, identifies the payer, and selects the participants. The Expense Splitting Module splits the amount equally or by a custom share, calculates the member balances, and updates the balances instantly. The Debt Simplification Module aggregates the net balances, minimizes the number of transactions, and suggests optimal settlements. Finally, the Settlement and Reporting Module tracks the payments made by members, updates the balances, and shows the pending dues.

The Admin Panel provides an oversight role in the architecture. It allows an administrator to view all groups, monitor transactions, and generate reports, which helps in maintaining the correctness of the data across the system. At the back end, the system relies on a centralized MySQL database that stores all essential information, including user details, groups, expenses, shares, balances, and settlement records. This centralized storage helps maintain accuracy and avoids duplication.

The data flow within the system follows a structured process in which the user input is validated, processed, and then stored in the database. For instance, when a user adds an expense, the system validates the amount and the selected participants, calculates the share of each participant, and updates the net balances of the group. When the user opens the debt simplification screen, the stored balances are read and a minimal set of settlement transactions is generated. After a payment is recorded, the balances are updated and the result is shown in the final expense and settlement

report. Overall, the architecture integrates the different modules into a unified framework and ensures a smooth flow of information from the input to the final output.

CHAPTER 6

MODULE DESCRIPTION

6.1 AUTHENTICATION MODULE

The Authentication Module is responsible for ensuring that only registered users can access the system. It manages user registration, login, and logout processes. During registration, the user provides basic details such as name and email address along with a password, and the system validates these details before creating the account.

During login, the credentials entered by the user are verified against the stored records. After a successful login, the user is taken to the home page, from where the groups and expenses can be accessed. This module forms the entry point of the system and ensures that financial information of a group is displayed only to the authorized members.

6.2 USER AND GROUP MANAGEMENT MODULE

The User and Group Management Module organizes the people who participate in shared spending. It allows users to create a new group, give it a name and description, and add members to it. The module stores the user details and assigns members to the groups, so that every expense can later be linked to the correct set of people.

The group screen presents the available groups along with their members and provides options such as managing members and viewing balances. Examples of groups include classmates sharing food costs, friends travelling together, or roommates sharing household bills. This module provides the foundation for all other modules because expenses and balances must always be associated with the correct group members.

6.3 EXPENSE ENTRY MODULE

The Expense Entry Module records the details of a shared purchase or payment. The user selects the group, enters a description of the expense, chooses a category, enters the total amount, selects the member who paid, and specifies the date of the expense. The participants who share the cost are then selected from the members of the group.

The module validates the information before it is stored, such as checking that the amount is a valid positive number and that at least one participant has been selected. Participant selection is important because not every member necessarily shares every expense. The recorded expense is passed to the expense splitting module for share calculation.

6.4 EXPENSE SPLITTING MODULE

The Expense Splitting Module calculates how much of an expense belongs to each selected participant. It supports three methods: equal split, percentage-based split, and custom split. In an equal split, the total amount is divided evenly among the selected participants. In a percentage-based split, each participant is assigned a percentage of the total, and the percentages must add up to one hundred. In a custom split, an individual amount is entered for each participant, and these amounts must add up to the total expense.

The module separates the person who paid the expense from the people who are responsible for sharing it, which is necessary for accurate balance calculation. For example, if an expense of ₹1,200 is shared equally by four members, each member’s share is ₹300, including the share of the payer when the payer is also a participant.

6.5 BALANCE CALCULATION MODULE

The Balance Calculation Module determines the net balance of each member of the group. After the shares are calculated, the module compares the amount each member has paid with the amount that member is responsible for. A positive net balance indicates that the member should receive money, while a negative net balance indicates that the member should pay money.

The balances are updated instantly whenever a new expense or payment is recorded. For a correctly recorded group, the sum of all net balances is zero, apart from rounding differences. The balances calculated by this module are the input to the debt simplification module and are also displayed to the members on the group screens.

6.6 DEBT SIMPLIFICATION MODULE

The Debt Simplification Module is the key computational feature of the project. Instead of asking every member to settle every expense separately, it combines the individual net balances and recommends a smaller set of transfers that clears the same overall obligations. Members with negative balances are treated as debtors, members with positive balances are treated as creditors, and members whose balances are zero need not make or receive any payment.

The module applies a greedy strategy. It repeatedly selects a debtor with an outstanding amount and a creditor who is owed money and recommends a transfer equal to the smaller of the two amounts. That amount is deducted from both balances, and the process repeats until all balances are cleared. Each transfer clears at least one member, so the number of suggested transfers never exceeds one less than the number of members involved.

The output of the module is a list of suggested settlements, each showing the member who should pay, the member who should receive the money, and the

amount. These are recommendations derived from the net balances and are not treated as completed payments until they are recorded.

6.7 EXPENSE HISTORY MODULE

The Expense History Module maintains a complete record of all expenses entered in a group. The expenses are displayed in a table showing the date, description, category, the member who paid, the split method, the amount, and the number of participants. A search option and a category filter help users locate a particular entry quickly.

This module allows users to review recorded entries and check details rather than relying on memory or separate notes. It also helps the members understand which records contribute to the current balances, and it provides a reliable reference whenever a question about an earlier expense arises.

6.8 SETTLEMENT AND REPORTING MODULE

The Settlement and Reporting Module presents the result of the balance calculations in a form that group members can understand. It supports reviewing who owes whom, how much is involved, and whether outstanding dues remain. A settlement represents a payment made to clear an outstanding obligation, and the module distinguishes between a suggested transaction and a payment that has been recorded.

After a settlement is recorded, the relevant balances are updated consistently. The final expense and settlement report summarizes the group’s financial position, including the total expenses, the total payments settled, the accounting state, and a table of member balances with their current status. Pending status helps users identify obligations that have not yet been cleared.

6.9 ADMINISTRATIVE OVERSIGHT MODULE

The Administrative Oversight Module provides tools for monitoring and managing the operations of the system. Through the Admin Panel, an administrator can view all the groups created in the system, monitor the transactions recorded by the groups, and generate reports. This ensures that the activities within the platform are properly tracked and controlled.

By providing a structured overview of all groups and transactions, the module makes it easier to detect irregularities or inconsistencies and to support the users when a problem occurs. Overall, this module helps in maintaining the stability and reliability of the entire system.

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The Intelligent Settlement System provides an integrated platform that simplifies the management of shared expenses by combining multiple features such as group and member management, expense recording, flexible expense splitting, net balance calculation, debt simplification, and settlement reporting. By using a debt simplification algorithm, it reduces the number of transfers required to clear the dues of a group and replaces confusing manual calculations with clear and consistent results. The expense history and the final expense and settlement report give the members a transparent record of every transaction.

The system is developed as a user-friendly website using HTML, CSS, and JavaScript for the front end, PHP for the back end, and MySQL for data storage. Its modular structure makes the main functions easy to understand, test, and maintain, and it provides a strong foundation for further development. The primary contribution of the project is the transformation of scattered shared expenses into a consolidated and actionable settlement plan that saves time, avoids disputes, and improves financial coordination within a group.

7.2 FUTURE ENHANCEMENT

The Intelligent Settlement System can be enhanced with additional features to make it smarter and more convenient. Automated due notifications can be added to send reminders for outstanding balances through e-mail or messages, so that pending dues are cleared on time. Integration with online payment gateways can allow members to complete the suggested settlements directly through the system, and the payment status can then be updated automatically.

Further improvements include exportable reports in PDF and spreadsheet formats, support for multiple currencies, and additional split rules such as weighted shares and item-level splitting. Receipt scanning can be introduced to fill in expense details automatically from a photograph of a bill. An analytics dashboard can show spending patterns by category and by member, and a mobile application can make it easier to record expenses on the move. The interface can also be improved with role-based access control, better accessibility, and unit testing of the splitting and settlement logic to increase reliability.

APPENDIX

A. SAMPLE CODE

```
import java.util.*;

class User {
    int id;
    String name;
    double balance;

    User(int id, String name) {
        this.id = id;
        this.name = name;
        this.balance = 0;
    }
}

class Expense {
    String category;
    String date;
    String payer;
    double amount;
    String status;

    Expense(String category, String date,
            String payer, double amount) {

        this.category = category;
        this.date = date;
        this.payer = payer;
        this.amount = amount;
        this.status = "Pending";
    }
}

public class FULLCODE {

    static ArrayList<User> users = new ArrayList<>();
    static ArrayList<Expense> expenses = new ArrayList<>();

    // MODULE 1 - USER & GROUP MANAGEMENT
    // MODULE 2 - EXPENSE ENTRY
    // (input handling omitted - see full program)

    // MODULE 3 - EXPENSE SPLITTING
```

```
static void splitExpense(
        User payer,
        ArrayList<User> participants,
        double amount) {

    System.out.println("\nSelect Split Method:");
    System.out.println("1. Equal Split");
    System.out.println("2. Percentage Split");
    System.out.println("3. Custom Split");

    System.out.print("Enter choice: ");
    int choice = sc.nextInt();

    double[] shares =
        new double[participants.size()];

    // EQUAL SPLIT

    if (choice == 1) {

        double share =
            amount / participants.size();

        for (int i = 0;
             i < participants.size();
             i++) {

            shares[i] = share;
        }
    }

    // PERCENTAGE SPLIT

    else if (choice == 2) {

        double totalPercentage = 0;

        for (int i = 0;
             i < participants.size();
             i++) {

            System.out.print(
                "Enter percentage for " +
                participants.get(i).name +
                ": "
            );

            double percentage =
```

```
                sc.nextDouble();

            shares[i] =
                amount * percentage / 100;

            totalPercentage += percentage;
        }

        if (Math.abs(
                totalPercentage - 100) > 0.01) {

            System.out.println(
                "Invalid percentage."
            );

            return;
        }
    }

    // CUSTOM SPLIT

    else if (choice == 3) {

        double total = 0;

        for (int i = 0;
             i < participants.size();
             i++) {

            System.out.print(
                "Enter share for " +
                participants.get(i).name +
                ": ₹"
            );

            shares[i] = sc.nextDouble();

            total += shares[i];
        }

        if (Math.abs(
                total - amount) > 0.01) {

            System.out.println(
                "Invalid shares."
            );

            return;
```

```
            }
        }

        else {

            System.out.println(
                "Invalid choice."
            );

            return;
        }

        // UPDATE BALANCES

        payer.balance += amount;

        System.out.println(
            "\nExpense Split:"
        );

        for (int i = 0;
             i < participants.size();
             i++) {

            participants.get(i).balance
                -= shares[i];

            System.out.println(
                participants.get(i).name +
                " : ₹" +
                String.format(
                    "%.2f",
                    shares[i]
                )
            );
        }
    }

    // MODULE 4 - DEBT SIMPLIFICATION

    static void simplifyDebts() {

        ArrayList<User> debtors =
            new ArrayList<>();

        ArrayList<User> creditors =
            new ArrayList<>();
```

```
for (User user : users) {

    if (user.balance < -0.01) {

        debtors.add(user);
    }

    else if (user.balance > 0.01) {

        creditors.add(user);
    }
}

System.out.println(
    "\nSIMPLIFIED SETTLEMENT"
);

System.out.println(
    "------------------------"
);

int i = 0;
int j = 0;

while (i < debtors.size() &&
       j < creditors.size()) {

    User debtor =
        debtors.get(i);

    User creditor =
        creditors.get(j);

    double amount =
        Math.min(
            -debtor.balance,
            creditor.balance
        );

    System.out.println(
        debtor.name +
        " pays " +
        creditor.name +
        " : ₹" +
        String.format(
            "%.2f",
            amount
        )
```

```
            );

            debtor.balance += amount;

            creditor.balance -= amount;

            if (Math.abs(
                    debtor.balance) < 0.01) {

                debtor.balance = 0;

                i++;
            }

            if (Math.abs(
                    creditor.balance) < 0.01) {

                creditor.balance = 0;

                j++;
            }
        }
    }

    // MODULE 5 - SETTLEMENT & REPORTING
    // (status prompts and final report omitted - see full program)

    public static void main(String[] args) {

        System.out.println(
            "==================================="
        );

        System.out.println(
            "       EXPENSE SPLITTER"
        );

        System.out.println(
            "==================================="
        );

        // MODULE 1

        createUsers();

        // MODULE 2 & 3

        System.out.print(
```

```
            "\nEnter number of expenses: "
        );

        int numberOfExpenses =
            sc.nextInt();

        sc.nextLine();

        for (int i = 1;
             i <= numberOfExpenses;
             i++) {

            System.out.println(
                "\n========== EXPENSE " +
                i +
                " ==========");
            );

            addExpense();
        }

        // DISPLAY BALANCES

        displayBalances();

        // MODULE 4

        simplifyDebts();

        // MODULE 5

        settlementReport();

        sc.close();
    }
}
```

B. SCREENSHOTS

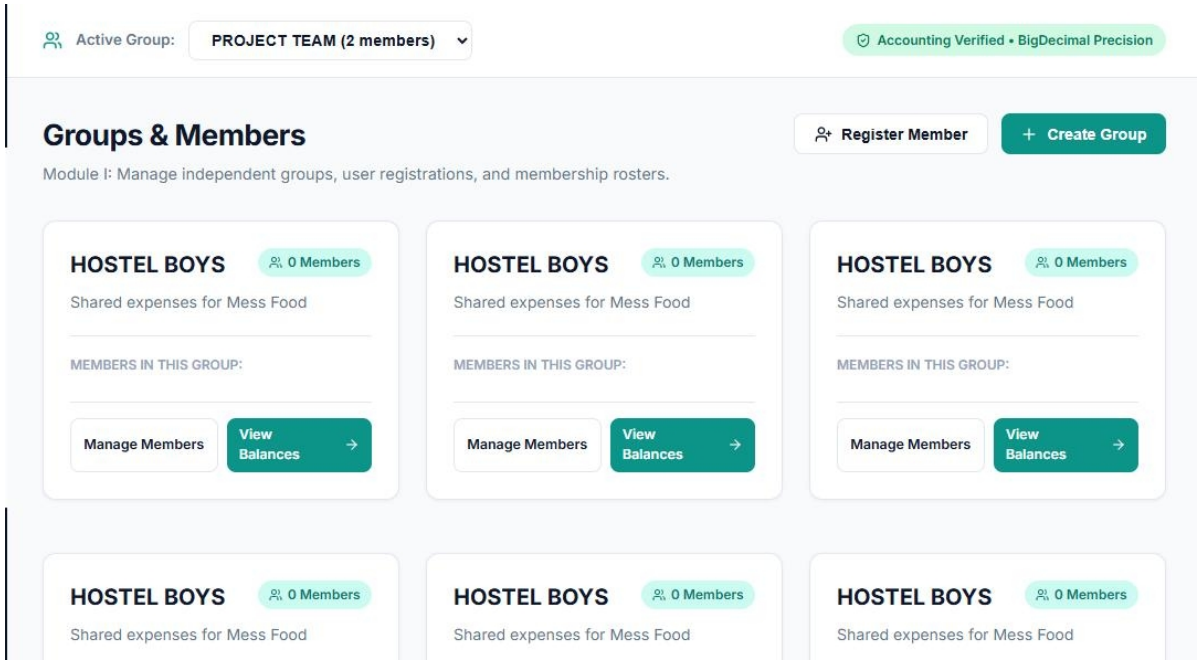


Fig.No. B.1 Group and Member Management

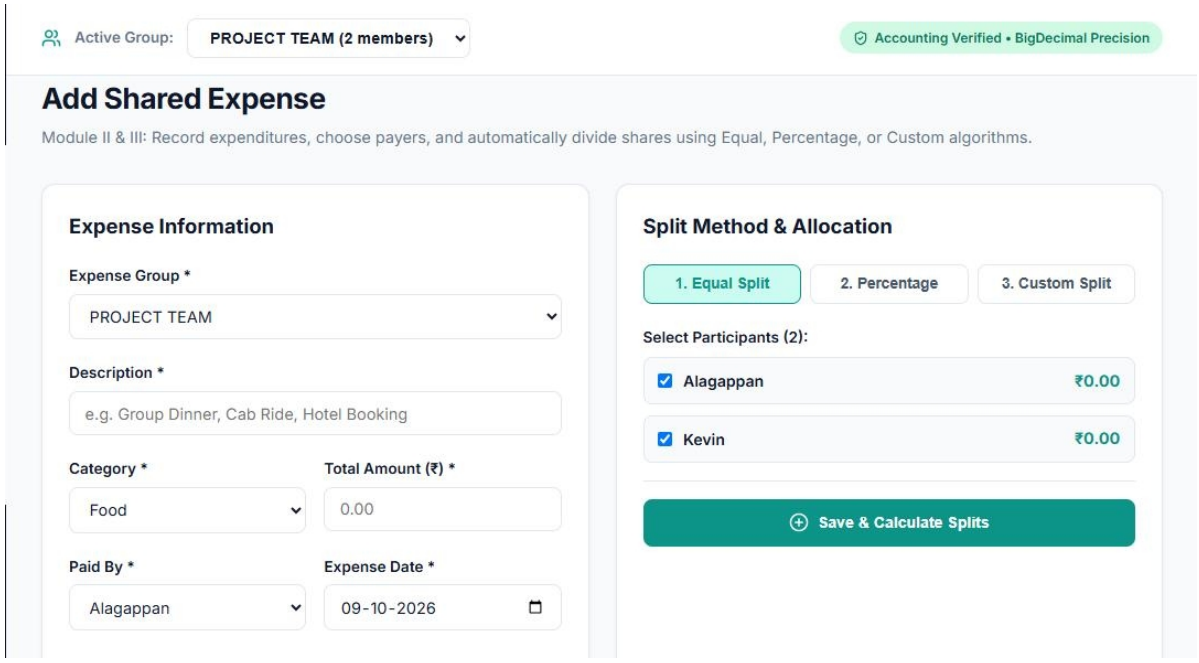


Fig.No. B.2 Add Shared Expense

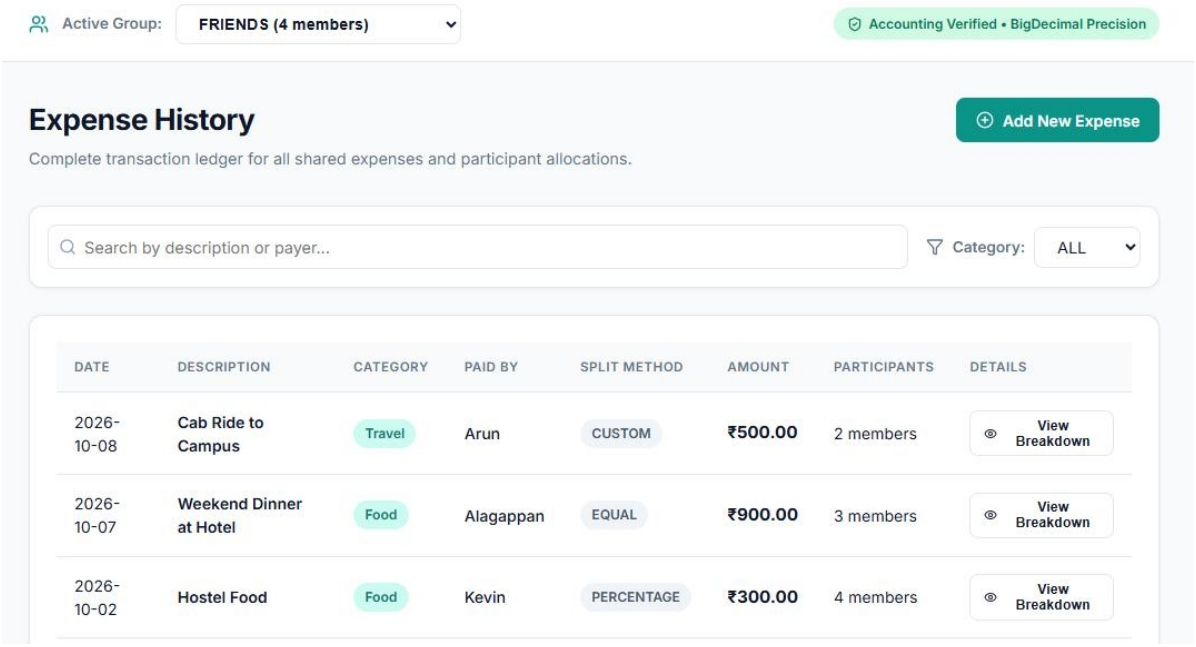


Fig.No. B.3 Expense History

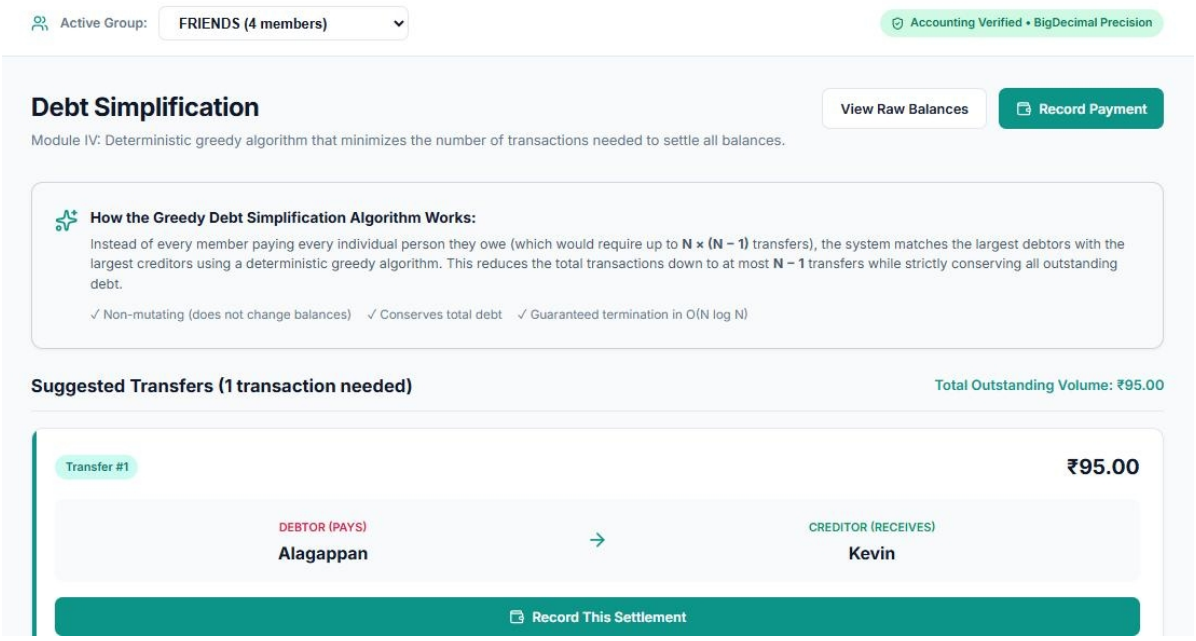


Fig.No. B.4 Debt Simplification

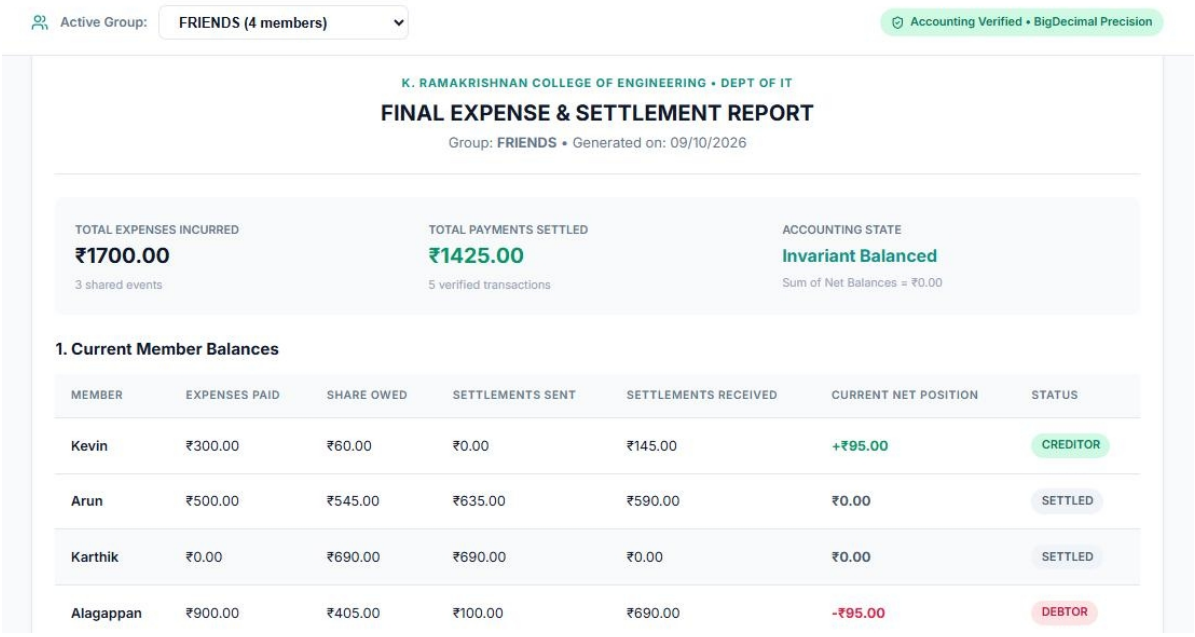


Fig.No. B.5 Final Expense and Settlement Report

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